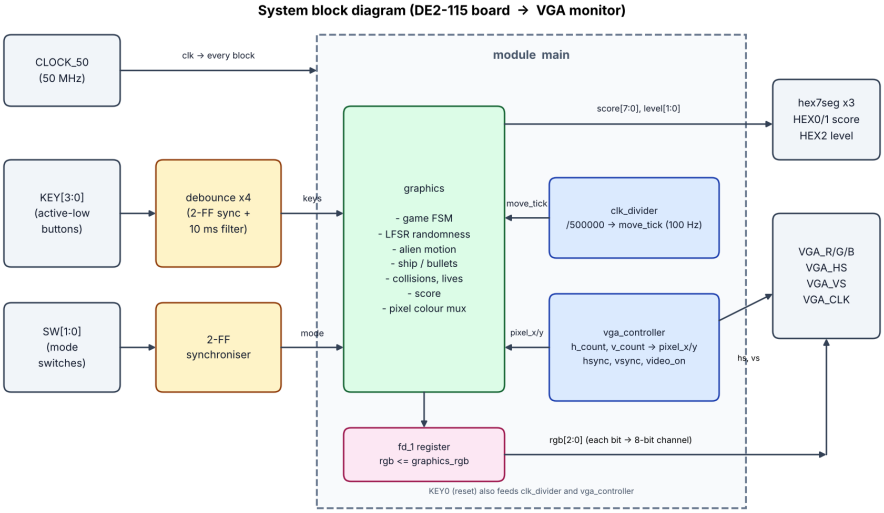
# Space Invaders on the DE2-115: modules and how they interact




## What each module does

- **de2115_top (board wrapper):** connects the design to the board pins. Cleans up keys and switches, turns the 3-bit colour into the VGA pins, drives the 7-segment displays.
- **debounce (x4, one per KEY):** removes contact chatter and synchronises the button to the 50 MHz clock, so the game sees one clean press.
- **Switch synchroniser (SW[1:0]):** two flip-flops per switch to make the asynchronous switch safe to read. No debounce needed.
- **main:** the core. Connects the VGA controller, the tick divider and the game, and holds the output register on the colour.
- **clk_divider:** counts 500,000 clocks and gives one pulse every 10 ms (move_tick). All game motion advances on that pulse.
- **vga_controller:** counts the raster position (pixel_x, pixel_y), makes the sync pulses and video_on.
- **graphics:** the whole game: title/play/over/won state machine, alien motion, ship and bullets, collisions, lives, score, random numbers, and the colour of each pixel.
- **hex7seg (x3):** turns a hex digit into 7-segment patterns for HEX0 to HEX2.

## How they interact

- **KEY0:** through debounce it becomes not_reset. It resets the VGA controller, the divider and the game, and sends the game back to the title screen.
- **KEY1 / KEY2 / KEY3:** through debounce, inverted so 1 = pressed, they become right, left and shoot, going into graphics. On the title screen any of them starts the game.
- **SW[1:0]:** through the synchroniser they become mode, going into graphics. On the title screen it sets the text MODE EASY/MEDIUM/HARD; when a key starts the game the mode is latched and stays fixed until the next reset.
- **clk_divider to graphics:** move_tick tells the game when to take its next 10 ms step.
- **vga_controller to graphics:** pixel_x, pixel_y and video_on say which pixel is being painted now; the game answers with the colour for exactly that pixel.
- **graphics to output register to VGA pins:** the 3-bit colour passes through one flip-flop and each bit is stretched to 8 bits for R, G and B.
- **vga_controller to monitor:** hsync and vsync go straight to the VGA pins so the monitor knows where each line and frame starts.
- **graphics to hex7seg:** score goes to HEX0/HEX1 and level to HEX2.
- **Clock:** CLOCK_50 feeds every module. The slower rates are enable pulses, not separate clocks.